

Örnek: $8,04 - 3,325 = ?$

$$\begin{array}{r} 12,3\cancel{7} \\ + 2,0\cancel{84} \\ \hline \downarrow \quad \downarrow \\ 14,454 \quad \checkmark \end{array}$$

Örnek: $12,37 + 2,084 = ?$

$$\begin{array}{r} 713 \\ 8,0410 \\ - 3,328 \\ \hline 4,713 \quad \checkmark \end{array}$$

Örnek: $36,5 \cdot 0,24 = ?$

$$\begin{array}{r} 36,5 \\ \times 0,24 \\ \hline 1460 \\ 730 \\ \hline 8,760 \end{array}$$

Örnek:

$$\left[\left(-\frac{3}{4} \right) : \frac{2}{3} \right] \cdot \left[\frac{5}{3} \cdot \left(-\frac{4}{3} \right) \right] =$$

$$\left(-\frac{3}{4} \right) \cdot \frac{3}{2}$$

$$-\frac{9}{8} \cdot -\frac{20}{9} \Rightarrow \frac{20}{8} \Rightarrow \frac{5}{2}$$

Örnek: $\frac{\left(\frac{1}{5} : \frac{1}{3} \right) + \left(\frac{3}{4} - \frac{1}{8} \right)}{\frac{7}{8} : \frac{5}{7} \Rightarrow \frac{7}{8} \cdot \frac{7}{5}} =$

$$\Rightarrow \frac{\frac{3}{5} + \frac{5}{8}}{\frac{49}{40}} = \frac{\frac{24+25}{40}}{\frac{49}{40}} \Rightarrow \frac{49}{40}$$

$$\frac{49}{40} \cdot \frac{40}{49} = 1$$

Örnek:

$$\left(\frac{1-\frac{1}{3}}{1+\frac{1}{3}} \right)^2 \cdot \left(\frac{7}{2} \right)^2 =$$

$$\frac{49}{4}$$

$$\Rightarrow \left(\frac{-\frac{1}{3}}{-\frac{2}{3}} \right)^2 \cdot \frac{49}{4} \Rightarrow \left(\frac{1}{3} \cdot \frac{3}{2} \right)^2 \cdot \frac{49}{4} = \left(\frac{1}{2} \right)^2 \cdot \frac{49}{4}$$

$$\frac{1}{4} \cdot \frac{49}{4} = \frac{49}{16}$$

Örnek:

$$1 + \frac{1}{1 + \frac{1}{1 + \frac{1}{3}}} = 1 + \frac{1}{1 + \frac{1}{1 + \frac{1}{3}}} = 1 + \frac{1}{1 + \frac{3}{4}} = 1 + \frac{1}{\frac{7}{4}} = 1 + \frac{4}{7} = \frac{11}{7}$$

Örnek:

$$1 + \frac{\frac{1}{2} + \frac{2}{3}}{1 + \frac{1}{1 + \frac{1}{2}}} = 1 + \frac{\frac{7}{6}}{1 + \frac{1}{\frac{3}{2}}} = 1 + \frac{\frac{7}{6}}{1 + \frac{2}{3}} = 1 + \frac{\frac{7}{6}}{\frac{5}{3}} = 1 + \frac{7}{6} \cdot \frac{3}{5} = 1 + \frac{7}{10} = \frac{17}{10}$$

Örnek:

$$\frac{17}{2} = 8 \frac{1}{2}$$

Örnek:

$$2 \frac{1}{5} = \frac{5 \cdot 2 + 1}{5} = \frac{11}{5}$$

$$-\frac{14}{9} = -1 \frac{5}{9}$$

$$-5 \frac{1}{3} = \frac{5 \cdot 3 + 1}{3} = -\frac{16}{3}$$

Örnek:

$$\frac{5}{6} + \frac{4}{5} + \frac{2}{3} =$$

(5)(6)(10)

$$\frac{25}{30} + \frac{24}{30} + \frac{20}{30} = \frac{69}{30} = 2 \frac{3}{10}$$

$$5 \frac{1}{2} - \frac{3}{5} =$$

$$\frac{11}{2} - \frac{3}{5} \Rightarrow \frac{55}{10} - \frac{6}{10} = \frac{49}{10} \Rightarrow 4 \frac{9}{10}$$

Örnek:

$$\frac{5}{8} \cdot \frac{3}{4} = \frac{15}{32}$$

$$\frac{2}{3} \cdot \frac{4}{5} = \frac{2}{3} \cdot \frac{29}{5} = \frac{58}{15} = 3 \frac{8}{15}$$

$$\text{Örnek: } 2 \frac{3}{4} \cdot (-3 \frac{5}{6}) \cdot \frac{24}{11} = ?$$

$$\frac{11}{4} \cdot (-\frac{23}{2}) \cdot \frac{24}{11} = -23$$

Örnek:

$$\frac{3}{4} : \frac{2}{3} = \frac{3}{4} \cdot \frac{3}{2} \Rightarrow \frac{9}{8}$$

$$2 \frac{1}{5} : 1 \frac{1}{3} = \frac{11}{5} : \frac{4}{3} \Rightarrow \frac{11}{5} \cdot \frac{3}{4} = \frac{33}{20}$$

Örnek:

$$\left(\frac{1}{2} + \frac{1}{4} \right) - \left(\frac{1}{3} - \frac{1}{4} \right) =$$

$$\left(\frac{7}{4} + \frac{5}{4} \right) - \left(\frac{3}{12} - \frac{3}{12} \right)$$

$$\left(\frac{28}{12} + \frac{15}{12} \right) - \left(\frac{6-5}{12} \right)$$

$$\frac{43}{12} - \frac{1}{12} \Rightarrow \frac{42}{12} = \frac{7}{2} \Rightarrow \frac{10}{3}$$

Örnekler:

1. $3 - 5 \cdot [2 - (1 - 8) \cdot 4 - 3 \cdot (-5)] - 2 - 5 \cdot (3 - 4) = ?$

$$3 - 5(2 + 28 + 15) - 2 + 5$$

$$3 - 5(45) - 2 + 5$$

$$3 - 225 - 2 + 5$$

$$3 - 225 + 3$$

$$\boxed{219} \checkmark$$

2. $(-3)^2 - [(-30) : 3 + 2 \cdot (-5) - 8 \cdot (2 - 5)] \cdot 2 - 9 = ?$

$$9 - (-10) + (-10) - (-24)$$

$$+ 24$$

$$9 - (-20) + 24 \cdot 2 - 9 - 3 \cdot (-X - 3) - 2X - 9$$

$$9 - (-4) \cdot 2 - 9$$

$$9 - (-8) - 9$$

$$-8$$

3. $x - \{1 - [2x - 3 \cdot (x + 1)]\} + 4 \cdot (2 - 2x) = ?$

$$2x - (3x + 3) + (8 - 8x)$$

$$1 - (2x - 3x - 3) + (8 - 8x)$$

$$1 - (-X - 3)$$

$$x + \{1 + X + 3\} + 8 - 8x$$

$$X - X - 4 + 8 - 8x$$

$$-8x + 4$$

4. $-3 \cdot [(x + 2y) - 3 - 2 \cdot (x + y)] - [(3x + 1) - (x - 8)] = ?$

$$x + 2y - 3 - (2x + 2y) - [3x + 1 - x + 8]$$

$$-3 \cdot (x + 2y - 3 - 2x - 2y) - (2x + 9)$$

$$-3 \cdot (-X - 3) - 2X - 9$$

$$-3 \cdot (-X - 3) - 2X - 9$$

$$3x + 9 - 2x - 9$$

$$x$$